AS · Unit 2 · WPH12 Waves & Electricity

Particle Nature of Light

Is light a wave or a stream of particles? Physicists argued about it for two hundred years, and the answer turned out to be "both, depending on the experiment". On this page you'll see why the photoelectric effect forced us to think of light as photons, how atoms emit and absorb light at exact energies, and how electrons, which we think of as particles, can diffract like waves.

Two models of light

A **model** in physics is a picture that lets you predict what an experiment will do. For electromagnetic (EM) radiation we have two, and you need to know what each one explains.

The wave model

- Light is an **electromagnetic wave**: oscillating electric and magnetic fields travelling at $c = 3.00 \times 10^{8}\ \text{m s}^{-1}$ in a vacuum.
- Energy is spread **continuously** over the wavefront. A brighter beam has a bigger amplitude.
- Explains diffraction, interference and polarisation.

The photon model

- Light is a stream of **photons**: separate packets (quanta) of energy.
- Each photon has energy $E = hf$ and gives **all** of it to one particle when it is absorbed.
- A brighter beam has **more photons per second**, not bigger photons.
- Explains the photoelectric effect and atomic line spectra.

How the models developed

The spec wants you to understand how our picture of light changed over time. The pattern is always the same: an experiment produces a result the current model can't explain, someone proposes a new model, and other scientists test it until it is accepted.

When	Who	What happened	Supports
1690	Huygens	Light is a wave; every point on a wavefront acts as a source of wavelets. Explains reflection and refraction, with light **slowing down** in glass.	wave
1704	Newton	Light is a stream of tiny "corpuscles". Explains straight-line travel and reflection, but predicts light **speeds up** in glass.	particle
1801	Young	Two-slit interference: bright and dark fringes. Two particles can't cancel each other out, but two waves can.	wave
1850	Foucault, Fizeau	Light is measured to travel more slowly in water than in air, as the wave model predicted. Newton's corpuscles are abandoned.	wave
1860s	Maxwell	Light is an electromagnetic wave; its speed can be calculated from electric and magnetic constants.	wave
1887	Hertz	Ultraviolet light helps sparks jump across a gap: the first sign of the photoelectric effect.	a puzzle

1900	Planck	To explain the radiation from hot objects, assumes energy is emitted in packets of size hf.	quanta
1902	Lenard	The energy of electrons released by light depends on its frequency, not on its intensity. The wave model can't explain this.	a puzzle
1905	Einstein	Light itself travels as photons with $E = hf$. Explains every feature of the photoelectric effect.	photon
1913	Bohr	Electrons in atoms have discrete energy levels, which explains the line spectrum of hydrogen.	photon
1916	Millikan	Careful experiments confirm Einstein's photoelectric equation and measure h.	photon
1923	Compton NOT IN SPEC	X-ray photons bounce off electrons like snooker balls, so photons carry momentum.	photon
1924	de Broglie	If waves can behave like particles, perhaps particles behave like waves, with $\lambda = h/p$.	both
1927	Davisson & Germer; G. P. Thomson	Electrons are diffracted by crystals, with exactly the wavelength de Broglie predicted.	both

KEY IDEA

Wave–particle duality

Neither model is "wrong". EM radiation behaves like a **wave** when it travels (diffraction, interference, polarisation) and like a stream of **particles** when it is emitted or absorbed (photoelectric effect, line spectra). Electrons do the same thing the other way round. We use whichever model explains the experiment in front of us.

Photons, E = hf and the electronvolt

DEFINITION

A **photon** is a quantum (a discrete packet) of electromagnetic radiation. Its energy is proportional to the frequency of the radiation: $E = hf$.

Photon energy

FORMULA SHEET

$$E = hf$$

E: energy of one photon / J $h = 6.63 \times 10^{-34}$ J s

f: frequency / Hz

Using $c = f\lambda$

$$E = \frac{hc}{\lambda}$$

λ: wavelength / m $c = 3.00 \times 10^{8}$ m s^{-1}

Higher frequency means a more energetic photon. That's why ultraviolet light can damage skin cells while red light, however bright, can't: each UV photon carries enough energy to break a chemical bond, and each red photon doesn't. h is tiny, so one visible photon carries only about 4×10^{-19} J. A laser pointer sends out thousands of millions of millions of them every second, which is why light looks smooth and continuous to us.

In the photon model, the power of a beam is the number of photons per second multiplied by the energy of each one:

$$P = N\,hf \quad \Rightarrow \quad N = \frac{P}{hf}$$

where N is the number of photons per second. At the same power, a beam of short-wavelength photons contains *fewer* photons, because each one carries more energy.

WORKED EXAMPLE

A laser pointer emits red light of wavelength 633 nm with a power of 1.0 mW. Calculate (a) the energy of one photon and (b) the number of photons emitted per second.

Show solution

(a) Use $E = hc/\lambda$, with the wavelength in metres:

$$E = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{633 \times 10^{-9}} = 3.14 \times 10^{-19}\ \text{J}$$

(b) Power is energy per second, so divide it by the energy of one photon:

$$N = \frac{P}{E} = \frac{1.0 \times 10^{-3}}{3.14 \times 10^{-19}} = 3.2 \times 10^{15}\ \text{photons per second}$$

The electronvolt

Joules are a clumsy size for single photons and electrons. Instead we use the **electronvolt**. From $V = W/Q$, the energy transferred to a charge moving through a p.d. is $W = QV$. For one electron moving through 1 V:

$$W = (1.60 \times 10^{-19}\ \text{C}) \times (1\ \text{V}) = 1.60 \times 10^{-19}\ \text{J}$$

DEFINITION

One **electronvolt (1 eV)** is the energy transferred when an electron moves through a potential difference of 1 volt. $1\ \text{eV} = 1.60 \times 10^{-19}$ J.

- **eV → J:** multiply by 1.60×10^{-19}. **J → eV:** divide by 1.60×10^{-19}.
- The 633 nm photon above has $3.14 \times 10^{-19} \div 1.60 \times 10^{-19} = 1.96\ \text{eV}$. Visible photons are all between about 1.7 eV (red) and 3.3 eV (violet).
- An electron accelerated from rest through a p.d. of 2000 V gains 2000 eV of kinetic energy (3.20×10^{-16} J).

EXAM TIP

A quick check for photon energies: E in eV is about $1240 \div \lambda$ in nm (so 620 nm gives 2.0 eV). This comes from hc/e; you aren't given it, so use it to check your working, not instead of it.

COMMON MISTAKE

Never subtract an energy in eV from an energy in joules. If a work function is given in eV and you have worked out hf in joules, convert one of them first. Also convert nm to m ($\times 10^{-9}$) before using $E = hc/\lambda$.

The photoelectric effect

When EM radiation of high enough frequency shines on a clean metal surface, **electrons are emitted** from the surface. They are called **photoelectrons**. A classic demonstration puts a clean zinc plate on top of a negatively charged gold-leaf electroscope. Ultraviolet light from a mercury lamp makes the leaf fall straight away, because electrons are leaving the zinc. Bright visible light does nothing, however bright it is. A sheet of glass in front of the lamp (glass absorbs UV) stops the effect.

KEY IDEA

What experiments show

1. No electrons are emitted if the frequency is below a **threshold frequency** f_0, however intense the radiation.
2. Above f_0, electrons are emitted **immediately**, even when the radiation is very weak.
3. The photoelectrons have a range of kinetic energies up to a **maximum**. The maximum kinetic energy increases with frequency but does **not** depend on intensity.
4. Above f_0, the **number** of electrons emitted per second is proportional to the intensity.
5. f_0 is different for different metals.

Einstein's explanation

- The radiation arrives as photons, each with energy hf.
- **One photon interacts with one electron** and gives it all its energy. It's all or nothing.
- Electrons are held in the metal by the attraction of the positive ions. The least energy needed to get an electron out is the **work function** ϕ.
- If $hf < \phi$, the electron can't escape. It shares the energy with the metal as heat within a tiny fraction of a second, so it can't save up energy from a second photon.
- If $hf > \phi$, the electron escapes and keeps the leftover energy as kinetic energy. Electrons from below the surface lose some energy on the way out, so $hf - \phi$ is the **maximum** kinetic energy.

DEFINITION

The **work function** ϕ of a metal is the minimum energy needed to remove an electron from its surface.

The **threshold frequency** f_0 is the minimum frequency of electromagnetic radiation that will cause photoelectrons to be emitted from the surface of the metal. The two are linked by $\phi = hf_0$.

Einstein's equation

FORMULA SHEET

$$hf = \phi + \tfrac{1}{2}mv_{\max}^2$$

hf: energy of one photon / J ϕ: work function / J $\tfrac{1}{2}mv_{\max}^2$: maximum kinetic energy of a photoelectron / J

Where Einstein's photoelectric equation comes from

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. One photon is absorbed by one electron, so the energy the electron receives is the whole photon energy:

$$E_{\text{in}} = hf$$

2. To leave the metal, the electron must do work against the attraction of the positive ions. The least work any electron has to do is the work function ϕ, for an electron right at the surface.

$$W_{\text{escape}} \geq \phi$$

3. Energy is conserved, so whatever is left over becomes kinetic energy:

$$\tfrac{1}{2}mv^2 = hf - W_{\text{escape}} \leq hf - \phi$$

This is why the electrons come out with a spread of speeds: different electrons lose different amounts of energy on the way out.

4. The fastest electrons are the ones that only had to do work ϕ. Rearranging:

$$\boxed{hf = \phi + \tfrac{1}{2}mv_{\max}^2}$$

5. At the threshold frequency the photon has only just enough energy, so the maximum kinetic energy is zero:

$$hf_0 = \phi + 0 \quad \Rightarrow \quad \boxed{f_0 = \frac{\phi}{h}}$$

Below f_0, $hf - \phi$ would be negative: no electron can escape, however many photons arrive.

The graph of maximum kinetic energy against frequency

Rearrange Einstein's equation into the form $y = mx + c$:

$$\underbrace{\tfrac{1}{2}mv_{\max}^2}_{y} = \underbrace{h}_{m}\ \underbrace{f}_{x}\ \underbrace{-\phi}_{c}$$

- The **gradient is** h, the same for every metal, so different metals give **parallel lines**.
- The line crosses the frequency axis at the **threshold frequency** f_0.
- Extended back to $f = 0$, it meets the energy axis at $-\phi$.

WORKED EXAMPLE

Ultraviolet radiation of wavelength 250 nm falls on a zinc plate. The work function of zinc is 4.3 eV. Calculate (a) the maximum kinetic energy of the photoelectrons in J, (b) their maximum speed and (c) the threshold frequency for zinc.

Show solution

(a) Photon energy, and the work function in joules:

$$hf = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{250 \times 10^{-9}} = 7.96 \times 10^{-19}\ \text{J}$$

$$\phi = 4.3 \times 1.60 \times 10^{-19} = 6.88 \times 10^{-19}\ \text{J}$$

$$\tfrac{1}{2}mv_{\max}^2 = hf - \phi = 7.96 \times 10^{-19} - 6.88 \times 10^{-19} = 1.08 \times 10^{-19}\ \text{J}$$

(b) Rearrange the kinetic energy for v, using $m_e = 9.11 \times 10^{-31}$ kg:

$$v_{\max} = \sqrt{\frac{2 \times 1.08 \times 10^{-19}}{9.11 \times 10^{-31}}} = 4.9 \times 10^{5}\ \text{m s}^{-1}$$

(c) At threshold, $hf_0 = \phi$:

$$f_0 = \frac{\phi}{h} = \frac{6.88 \times 10^{-19}}{6.63 \times 10^{-34}} = 1.0 \times 10^{15}\ \text{Hz}$$

That's a wavelength of $c/f_0 \approx 290$ nm. Visible light (above 380 nm) can never release electrons from zinc.

Why the photoelectric effect needs photons

The spec asks you to explain how the photoelectric effect is evidence for the **particle nature** of EM radiation. The argument is that the wave model makes predictions that turn out to be wrong, and the photon model gets every one right.

Observation	Wave model predicts	Photon model explains
Threshold frequency	Energy arrives continuously, so any frequency should release electrons if the light is bright enough (or you wait long enough).	Each electron takes energy from one photon. If $hf < \phi$, no single photon has enough, and more photons per second just means more failures.
Emission is instant	In dim light an electron needs time to soak up enough energy, so there should be a delay.	Energy arrives in lumps. The first photon to hit can free an electron immediately.
Maximum KE depends on frequency, not intensity	Brighter light has a bigger amplitude, so it should shake electrons out faster.	$\tfrac{1}{2}mv_{\max}^2 = hf - \phi$ depends only on the energy of one photon. Brighter light means more photons, so more electrons, but not faster ones.

Number of electrons per second $\propto$ intensity	Fits.	Fits: twice as many photons per second free twice as many electrons.

KEY IDEA

How long would the wave model make you wait?

In the wave model an electron can only absorb the energy that falls on roughly the area of one atom, about $(10^{-10}\ \text{m})^2 = 10^{-20}\ \text{m}^2$. In very dim light of intensity $1.0 \times 10^{-4}\ \text{W m}^{-2}$, using $I = P/A$, the power reaching one atom is

$$P = IA = 1.0 \times 10^{-4} \times 10^{-20} = 10^{-24}\ \text{W}$$

To collect 2 eV (3.2×10^{-19} J) would take $3.2 \times 10^{-19} \div 10^{-24} \approx 3 \times 10^{5}$ s, about four days. In real experiments the first electrons appear within about 10^{-9} s.

EXAM TIP

A strong answer to "explain how the photoelectric effect provides evidence for the particle nature of light" makes these points: light arrives as photons with energy $E = hf$; **one photon is absorbed by one electron**; an electron needs at least the work function ϕ to escape; so below $f_0 = \phi/h$ no electrons are emitted **whatever the intensity**; in the wave model energy depends on intensity and arrives continuously, so it predicts emission at any frequency (after a delay).

⚠ Beyond the spec · not part of the Edexcel IAL specification

How the maximum kinetic energy is measured: stopping potential

In the lab, the photoelectrons are collected by a second electrode, and the photocurrent is measured with a sensitive ammeter. If you make the collecting electrode *negative*, it repels the electrons. Increase this reverse p.d. until the current just drops to zero: now even the fastest electrons are turned back. The p.d. that does this is the **stopping potential** V_s, and the work it does on the fastest electrons equals their kinetic energy:

$$eV_s = \tfrac{1}{2}mv_{\max}^2 = hf - \phi \quad \Rightarrow \quad V_s = \frac{h}{e}f - \frac{\phi}{e}$$

A graph of V_s against f is a straight line of gradient h/e. This is how Millikan measured h in 1916, and it is where the kinetic-energy values in data tables usually come from.

Energy levels and line spectra

Pass an electric current through a tube of hot hydrogen gas and look at the glow through a diffraction grating. Instead of a rainbow you see a few **sharp coloured lines**, always at the same wavelengths. Every element has its own pattern, like a fingerprint. Helium was discovered in 1868 from unknown lines in the Sun's spectrum before anyone found it on Earth.

The explanation is that an electron in an atom can only have certain **discrete energies**, called **energy levels**. It can move between levels, but it can never sit in between them.

DEFINITION

An **energy level** is one of the discrete values of energy that an electron in an atom is allowed to have. The lowest level is the **ground state**; higher levels are **excited states**.

The **ionisation energy** is the energy needed to remove an electron completely from an atom in its ground state.

- **Why the energies are negative.** Zero is defined as an electron that has just escaped from the atom. A bound electron has *less* energy than that, so its energy is negative. The ground state of hydrogen is $-13.6\ \text{eV}$, so its ionisation energy is $13.6\ \text{eV}$.
- **Emission.** When an electron drops from a higher level to a lower one, it emits **one photon** carrying exactly the energy difference.
- **Absorption.** An electron can jump up by absorbing a photon, but only if the photon's energy **exactly** matches the gap between two levels (or is enough to ionise the atom). Other photons pass straight through.
- **Collisions.** In a discharge tube, atoms are mostly excited by collisions with fast electrons. A colliding electron can hand over any amount of its kinetic energy, so it only needs *at least* the gap, and keeps the rest.

Photon from a transition

$$hf = E_1 - E_2$$

E_1: energy of the higher level / J E_2: energy of the lower level / J f: frequency of the photon / Hz

Because only certain energy *differences* exist, only certain photon frequencies can be emitted or absorbed. That is why the spectrum is a set of lines rather than a continuous rainbow.

- An **emission spectrum** (bright lines on a dark background) comes from a hot, low-pressure gas whose excited electrons are dropping down.
- An **absorption spectrum** (dark lines on a continuous spectrum) appears when white light passes through a cooler gas. Photons with exactly the right energies are absorbed, and when the excited electrons drop back, the photons are re-emitted in **all directions**, so very few continue in the original beam.
- The dark absorption lines are at **exactly the same wavelengths** as the bright emission lines of the same element, because they come from the same energy gaps.

WORKED EXAMPLE

An electron in a hydrogen atom drops from the level at $-1.51\ \text{eV}$ to the level at $-3.40\ \text{eV}$. Calculate the wavelength of the photon emitted and state the region of the EM spectrum it is in.

Show solution

Energy difference, converted to joules:

$$\Delta E = -1.51 - (-3.40) = 1.89\ \text{eV} = 1.89 \times 1.60 \times 10^{-19} = 3.02 \times 10^{-19}\ \text{J}$$

Frequency from $hf = \Delta E$, then wavelength from $c = f\lambda$:

$$f = \frac{3.02 \times 10^{-19}}{6.63 \times 10^{-34}} = 4.56 \times 10^{14}\ \text{Hz} \qquad \lambda = \frac{3.00 \times 10^{8}}{4.56 \times 10^{14}} = 6.58 \times 10^{-7}\ \text{m}$$

658 nm is **visible red light**. (The measured line is at 656 nm; the small difference comes from rounding the level energies to 3 significant figures.)

EXAM TIP

"How many different wavelengths can be emitted?" Count every possible jump downwards, not just the jumps to the ground state. An electron in level 4 can go 4→3, 4→2, 4→1, 3→2, 3→1 and 2→1: **six** lines. In general, n levels give $\frac{1}{2}n(n-1)$ lines.

COMMON MISTAKE

Hydrogen in its ground state will **not** absorb a 12.5 eV photon. The gaps are 10.2 eV (to level 2), 12.1 eV (to level 3) and 12.75 eV (to level 4), and 12.5 eV matches none of them. A 12.5 eV *electron*, on the other hand, can excite the atom to level 3 and carry on with 0.4 eV.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Where hydrogen's numbers come from: $E_n = -13.6\ \text{eV}/n^2$

For hydrogen only, the levels follow a simple pattern: $E_n = -\dfrac{13.6\ \text{eV}}{n^2}$, giving $-13.6, -3.40, -1.51, -0.85, \ldots$ eV. Bohr found this in 1913 by assuming the electron's orbit could only have certain sizes. De Broglie later gave a lovely reason: the orbit must fit a **whole number of electron wavelengths** around its circumference, $2\pi r = n\lambda$, so the electron's wave reinforces itself instead of cancelling out. The formula isn't in the spec, and it only works for hydrogen: for any atom with more than one electron you need the measured energy levels.

Electron diffraction: particles as waves

In 1924 de Broglie suggested that if light waves can behave like particles, then particles might behave like waves. He proposed that every moving particle has a wavelength that depends on its momentum:

de Broglie wavelength

FORMULA SHEET

$$\lambda = \frac{h}{p}$$

λ: wavelength / m $h = 6.63 \times 10^{-34}$ J s $p = mv$: momentum / kg m s⁻¹

More momentum means a shorter wavelength. To test the idea you need to show that electrons do something only waves can do, and the clearest test is **diffraction**. Diffraction is only noticeable when the gap is similar in size to the wavelength. For electrons accelerated through a few thousand volts, λ is about 10^{-11} m, which is close to the spacing between atoms in a crystal (about 10^{-10} m). So a thin crystal acts as a diffraction grating for electrons.

The electron diffraction tube

- A heated cathode releases electrons, which are accelerated through a p.d. V of a few kilovolts towards an anode with a hole in it.
- The beam passes through a very thin film of **graphite** (carbon). Its atoms are arranged in layers with regular spacings.
- On the fluorescent screen you see a bright central spot surrounded by **concentric rings**. The graphite is made of many tiny crystals pointing in random directions, so the diffracted beams spread out in cones, which hit the screen as rings.
- Particles travelling in straight lines would only make a single spot. The rings are a **diffraction pattern**, which is evidence that electrons have a **wave nature**.

- Increase V: the electrons gain more momentum, their wavelength gets **shorter**, they diffract less, and the rings get **smaller**. The measured ring sizes match $\lambda = h/p$.

Wavelength of an electron accelerated through a p.d. V

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 The electron starts at rest. Moving through a p.d. V, it gains energy $W = QV = eV$, all as kinetic energy:

$$\tfrac{1}{2}mv^2 = eV$$

2 Write the kinetic energy in terms of momentum. Since $p = mv$, $v = p/m$:

$$\tfrac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m} = eV$$

3 Solve for the momentum:

$$p = \sqrt{2meV}$$

4 Put this into the de Broglie equation:

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}}$$

So $\lambda \propto 1/\sqrt{V}$: four times the voltage halves the wavelength, and roughly halves the ring sizes.

WORKED EXAMPLE

Electrons are accelerated from rest through a p.d. of 4.0 kV. Calculate their de Broglie wavelength, and explain why a tennis ball of mass 0.058 kg moving at 30 m s^{-1} shows no diffraction effects.

Show solution

Momentum of the electron from $p = \sqrt{2meV}$:

$$p = \sqrt{2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 4000} = 3.42 \times 10^{-23}\ \text{kg m s}^{-1}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{3.42 \times 10^{-23}} = 1.94 \times 10^{-11}\ \text{m}$$

This is comparable to the spacing of atoms in a crystal, so the electrons diffract noticeably. For the tennis ball:

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.058 \times 30} = 3.8 \times 10^{-34}\ \text{m}$$

That is far smaller even than a proton (about 10^{-15} m across). There is no gap anywhere near that small, so the ball's diffraction is far too small ever to detect.

EXAM TIP

For "explain how electron diffraction shows electrons behave as waves", say: diffraction is a **wave** property; the electrons form a diffraction pattern (rings); this happens because their de Broglie wavelength is **similar to the spacing between atoms** in the graphite.

⚠ Beyond the spec · not part of the Edexcel IAL specification

One particle at a time

Turn a double-slit experiment down so low that only one photon (or one electron) is in the apparatus at a time. Each one lands at a single point on the detector, like a particle. But after thousands have arrived, the dots build up into interference fringes, like a wave. Each particle seems to pass through both slits and interfere with itself. G. I. Taylor first tried this idea in 1909, photographing a diffraction pattern with light so dim that the exposure took months, and in 1989 Akira Tonomura's team filmed single electrons building up interference fringes one dot at a time. The pattern tells you where a particle is *likely* to land, and this probability idea is the starting point of quantum mechanics.

Wave or particle? The evidence

Here is the whole topic in one table. "Wave" and "particle" are two models of the same thing, and each experiment shows you one side of it.

Observation	What it shows	Wave model	Particle model
Diffraction and interference of light	Light behaves as a wave	Explains it	Can't explain it
Polarisation of light	Light is a transverse wave	Explains it	Can't explain it
Photoelectric effect	Light is absorbed as photons	Fails (threshold, no delay)	Explains it
Atomic line spectra	Light is emitted as photons of fixed energy	Can't explain sharp lines	Explains it, with energy levels
Electron diffraction	Electrons behave as waves	Explains it, with $\lambda = h/p$	Predicts a single spot

KEY IDEA

Light and electrons both show **wave–particle duality**. A useful rule of thumb: wave behaviour shows up in how things **travel** and spread out; particle behaviour shows up when energy is **given or taken** in a single interaction.

Worked exam questions

WORKED EXAMPLE · GRAPH

A student measures the maximum kinetic energy of photoelectrons from a metal for several frequencies and draws a best-fit line. Two points on the line are $(7.0 \times 10^{14}\ \text{Hz},\ 0.62\ \text{eV})$ and $(1.10 \times 10^{15}\ \text{Hz},\ 2.28\ \text{eV})$. (a) Use the graph to find a value for the Planck constant. (b) Find the work function of the metal in eV. (c) Calculate the threshold frequency.

Show solution

(a) The gradient of the KE–f graph is h. Work it out in eV s, then convert to J s:

$$\text{gradient} = \frac{2.28 - 0.62}{1.10 \times 10^{15} - 7.0 \times 10^{14}} = \frac{1.66 \text{ eV}}{4.0 \times 10^{14} \text{ Hz}} = 4.15 \times 10^{-15} \text{ eV s}$$

$$h = 4.15 \times 10^{-15} \times 1.60 \times 10^{-19} = 6.64 \times 10^{-34} \text{ J s}$$

(b) Rearrange $\frac{1}{2}mv_{\max}^2 = hf - \phi$ and use either point:

$$\phi = hf - \tfrac{1}{2}mv_{\max}^2 = (4.15 \times 10^{-15} \times 7.0 \times 10^{14}) - 0.62 = 2.91 - 0.62 = 2.29 \text{ eV} \approx 2.3 \text{ eV}$$

(c) At threshold, $f_0 = \phi/h$:

$$f_0 = \frac{2.29 \text{ eV}}{4.15 \times 10^{-15} \text{ eV s}} = 5.5 \times 10^{14} \text{ Hz}$$

This is where the line would cross the frequency axis. The metal is probably sodium.

WORKED EXAMPLE · ENERGY LEVELS

Four of the energy levels of a mercury atom are -10.4 eV (ground state), -5.5 eV, -3.7 eV and -1.6 eV. (a) Calculate the wavelength of the photon emitted in the transition from -5.5 eV to the ground state, and name the region of the spectrum. (b) How many different photon energies could be emitted by atoms excited to the -1.6 eV level? (c) What is the ionisation energy of mercury in joules?

Show solution

(a) Energy gap in joules, then $\lambda = hc/\Delta E$:

$$\Delta E = -5.5 - (-10.4) = 4.9 \text{ eV} = 4.9 \times 1.60 \times 10^{-19} = 7.84 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{7.84 \times 10^{-19}} = 2.54 \times 10^{-7} \text{ m} = 254 \text{ nm}$$

This is **ultraviolet** (shorter than 380 nm). Fluorescent lamps use exactly this line: a coating on the inside of the tube absorbs the UV and emits visible light.

(b) There are four levels, so the number of possible downward jumps is $3 + 2 + 1 = 6$ different photon energies.

(c) Ionisation means going from the ground state to 0 eV: $10.4 \text{ eV} = 10.4 \times 1.60 \times 10^{-19} = 1.66 \times 10^{-18}$ J.

WORKED EXAMPLE · MULTI-STEP

In an electron diffraction tube, electrons are accelerated through 3.5 kV. (a) Show that their de Broglie wavelength is about 2×10^{-11} m. (b) The p.d. is increased to 5.0 kV. Explain what happens to the rings on the screen. (c) Calculate the speed a neutron (mass 1.67×10^{-27} kg) would need to have the same wavelength as in (a).

Show solution

(a) $\lambda = h/\sqrt{2meV}$:

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 3500}} = \frac{6.63 \times 10^{-34}}{3.19 \times 10^{-23}} = 2.08 \times 10^{-11}\ \text{m} \approx 2 \times 10^{-11}\ \text{m}$$

(b) The electrons gain more kinetic energy and momentum, so $\lambda = h/p$ decreases (to $2.08 \times 10^{-11} \times \sqrt{3.5/5.0} = 1.74 \times 10^{-11}$ m). Shorter waves diffract through smaller angles, so the rings get **smaller** (and closer to the central spot).

(c) Same wavelength means the same momentum, so $v = h/(m\lambda)$:

$$v = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times 2.08 \times 10^{-11}} = 1.9 \times 10^{4}\ \text{m s}^{-1}$$

The neutron is about 1800 times heavier than the electron, so it needs to go about 1800 times slower to have the same momentum.

Quick checks

QUICK CHECK

Light above the threshold frequency shines on a metal. The intensity is doubled but the frequency stays the same. What happens?

- The maximum kinetic energy of the photoelectrons doubles.
- Twice as many photoelectrons are emitted per second, with the same maximum kinetic energy.
- The threshold frequency halves.
- The photoelectrons are emitted after a shorter delay.

Doubling the intensity doubles the number of photons per second. Each photon still has the same energy hf, so each electron gets the same energy: more electrons, same $\frac{1}{2}mv_{\max}^2 = hf - \phi$. There is no delay to shorten; emission is already instant.

QUICK CHECK

What is the gradient of a graph of maximum kinetic energy of photoelectrons (in J) against frequency of the radiation?

- The work function ϕ
- The threshold frequency f_0
- The Planck constant h
- The electron charge e

$\frac{1}{2}mv_{\max}^2 = hf - \phi$ is $y = mx + c$ with gradient h. The y-intercept is $-\phi$ and the x-intercept is f_0. Every metal gives the same gradient.

QUICK CHECK

A photon has an energy of 3.0 eV. What is its energy in joules?

- 1.9×10^{19} J
- 4.8×10^{-19} J
- 5.3×10^{-20} J
- 2.0×10^{-34} J

eV → J means multiply by 1.60×10^{-19}: $3.0 \times 1.60 \times 10^{-19} = 4.8 \times 10^{-19}$ J. The first option is what you get by dividing instead of multiplying.

QUICK CHECK

The accelerating voltage in an electron diffraction tube is increased from 2.0 kV to 8.0 kV. By what factor does the de Broglie wavelength of the electrons change?

- It becomes 4 times longer.
- It becomes 4 times shorter.
- It becomes 2 times longer.
- It becomes 2 times shorter.

$\lambda = h/\sqrt{2meV}$, so $\lambda \propto 1/\sqrt{V}$. Four times the voltage gives $\sqrt{4} = 2$ times the momentum, so half the wavelength, and the rings shrink.

QUICK CHECK

The ground state of hydrogen is $-13.6\ \text{eV}$ and the next levels are -3.40, -1.51 and $-0.85\ \text{eV}$. Which photon could be absorbed by a hydrogen atom in its ground state?

- 5.1 eV
- 10.2 eV
- 11.0 eV
- 12.5 eV

A photon is only absorbed if its energy exactly matches a gap: $-3.40 - (-13.6) = 10.2\ \text{eV}$. The other gaps from the ground state are 12.1 eV and 12.75 eV, and none of the other options matches any of them (and none is enough to ionise the atom).

QUICK CHECK

Which observation can the wave model of light *not* explain?

- Light spreads out after passing through a narrow slit.
- Light can be polarised.
- Red light never releases electrons from zinc, however bright it is.
- More photoelectrons are emitted when the light is brighter.

In the wave model, brighter light delivers more energy, so eventually any frequency should release electrons. The threshold frequency only makes sense if energy comes in photons of hf, one per electron. Diffraction and polarisation are wave behaviour, and the number of electrons rising with brightness fits both models.